

Program: Diploma in Artificial Intelligence	
Course Code: 4389	Course Title: R Programming Lab
Semester: 4	Credits: 0
Course Category: Program Core course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Learn the techniques in R language.
- Familiar with data manipulation, analysis and visualization
- Understand control structures and R functions.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sound knowledge of Python programming	1008	Introduction to IT systems Lab	I
Sound knowledge of Python programming	3347	Python Programming Lab	III
Knowledge of data sets	4347	Machine Learning Lab	IV

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Basics of R programming	10	Applying
CO2	Data Manipulation and visualization	10	Applying
CO3	Control structures and functions and Text Analysis	10	Applying
CO4	Open ended experiment	15	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	2	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Gain fundamentals of R programming language		
M1.01	Introduction, Downloading and Installing R,	4	Applying
M1.02	Downloading and Installing RStudio IDE	2	Applying
M1.03	Variables, Data types , arithmetic operators, order of operators, Relational/logical operators, print() function	2	Applying

M1.04	R Objects: vectors, matrices, lists	2	Applying
CO2	Data Manipulation and visualization		
M2.01	Loading external data into R	3	Applying
M2.02	Data manipulation using functions like subset, filter and mutate	4	Applying
M2.03	Basic data visualization using ggplot2 package	3	Applying
	Lab Exam – I		

CO3	Control structures, Functions and Text analysis		
M3.01	Writing custom functions and understanding function arguments	2	Applying
M3.02	Applying functions to data using the apply family functions	2	Applying
M3.03	Perform basic text analysis tasks on given text using R	3	Applying
M3.04	Sentiment analysis using packages like tm and tidytext	3	Applying
CO4	Open ended experiment		
M4.01	Students work on a mini-project using R to solve a real-world problem. Project might involve data analysis, visualization, and potentially some machine learning	15	Applying
	Lab Exam -2		

**** For example -Sentiment Analysis of Movie Reviews**

Text / Reference

T/R	Book Title/Author
T1	Tanveer Siddiqui, U.S. Tiwary, “Natural Language Processing and Information Retrieval”, Oxford University Press, 2008.
R1	Daniel Jurafsky and James H Martin, “Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition”, 2 nd Edition, Prentice Hall, 2008.